v1.0Lepton Deep Dive_ One-Lepton Universe

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Lepton Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: `v1.0Lepton Deep Dive_ One-Lepton Universe V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Muon g-2 status	Preliminary FNAL result	FNAL 2023 final: 5.1σ (KNT18) or 1.5σ (BMW). V2.0 stores both predictions; BMW update critical	HIGH	Status changed: BMW lattice reduces tension substantially. V1.0 statement of confirmed anomaly premature
Lepton universality R(D*)	SM universality assumed	LHCb 2023: R(D*)=0.281 (SM: 0.254) — 1.8σ tension in tau vs mu. LFU parameter added	MEDIUM	Novel tension in lepton universality not in V1.0
NuFIT 5.3 oscillation params	Earlier fit version	NuFIT 5.3 2024: $\sin^2\theta_{12}$=0.307, $\sin^2\theta_{23}$=0.546, delta_CP=-107°. PMNS matrix updated	HIGH	Oscillation parameters in V1.0 are out of date; these drive neutrino lepton behavior
Tau g-2 measured	Not measured in V1.0	Belle II 2023: a_τ constrained to [-0.0025,+0.0055] — SM (1.177×10^-3) confirmed	LOW	First precision tau g-2 constraint; confirms SM completeness of lepton sector

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Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Muon, NuFIT
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Lepton Deep Dive_ One-Lepton Universe V1.0 archived | V2.0 is the active specification as of June 2026